

1. Trust region problem

$$\min_{u \in \mathbb{R}^n} \quad m(u) = g^T u + \frac{1}{2} u^T B u$$
subject to $\|u\| \leq \Delta$

Problem: $B = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \Delta = \frac{1}{2}$
 $m(u) = g^T u + \frac{1}{2} u^T B u$
 $u, v \in \mathbb{R}^2$

$\Rightarrow p = -g$ (steepest descent) $\Rightarrow \begin{pmatrix} -1 \\ -1 \end{pmatrix}$
 $\Rightarrow m(x+1) = g^T(x+1) + \frac{1}{2}(x+1)^T B(x+1)$
 $\Rightarrow -g^T x_1 + \frac{1}{2} g^T x_2 + 1$
 $\Rightarrow m'(1) = -g^T \frac{1}{2} + \frac{1}{2} g^T \frac{1}{2} = 0$
 $\Rightarrow x = \frac{1}{2} \frac{2g}{g^T g} = \frac{2 \cdot 1}{4 \cdot 1} = \frac{1}{2} \in \frac{1}{2}$
 $\Rightarrow v = \frac{1}{2} \begin{pmatrix} 1+1 \\ 1-1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$
 $\Rightarrow \sqrt{\frac{1}{2} \cdot 1} \leq \frac{1}{2}$
 $\Rightarrow \frac{1}{2} \leq \frac{1}{2} \checkmark$
 $\Rightarrow u^* = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix} = p^*$

Problem 2: $B = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}, g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \Delta = \sqrt{\frac{2}{3}}$
 $m(u) = g^T u + \frac{1}{2} u^T B u$
 $m'(u) = g + u^T B u = 0$
 $\Rightarrow u = -\frac{2}{g}$
 $\frac{1}{2} = \sqrt{\frac{2}{3} \frac{0}{2}}$
 $\Rightarrow u = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$
 $\Rightarrow u = \begin{pmatrix} -1 \\ 1 \end{pmatrix} = p^*$
 $\Rightarrow \sqrt{\frac{1}{2} \cdot \frac{1}{2}} = \frac{1}{2} \leq \frac{1}{2} \checkmark$
 $\Rightarrow p^* \text{ not with trust region}$
 $\Rightarrow \text{line search } p^* \text{ not } p^*$
 $\Rightarrow p(x) = p^* + t(p^*, p^*)$
subject to $\|p(x)\| = \Delta = \sqrt{\frac{2}{3}}$
 $t \in [0, 1]$

$\Rightarrow \|p(x)\| = \Delta = \sqrt{\frac{2}{3}}$
 $\Rightarrow \left[\begin{pmatrix} -1 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right]^T = \frac{2}{3}$
 $\Rightarrow \left[\frac{1}{2} - \frac{1}{2} \right]^T + 2 \left[\frac{1}{2} - \frac{1}{2} \right] + \frac{1}{2} \left[\begin{pmatrix} 1 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right]^T = \frac{2}{3}$
 $\Rightarrow \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} t^2 = \frac{2}{3}$
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 $\Rightarrow t = \frac{1}{3}$
 $\Rightarrow p^* = \begin{pmatrix} -\frac{2}{3} \\ \frac{1}{3} \end{pmatrix} + \frac{1}{3} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
 $= \begin{pmatrix} -\frac{1}{3} \\ \frac{2}{3} \end{pmatrix}$

Problem 3: $B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \Delta = 1$
 $L(u, \lambda) = g^T u + \frac{1}{2} u^T B u + \lambda(u+1)^T B u$
primal feasibility $u \in \Delta \Rightarrow u \in \mathbb{R}^2$
stationary $\nabla_u L = 0 \Rightarrow u^T B u = g$
dual feasibility $\lambda \geq 0$

Weak duality complementary $\lambda(u+1)^T B u = 0$
 $L(u, \lambda) = g^T u + \frac{1}{2} u^T B u + \lambda(u+1)^T B u$
Consider $\lambda = 0$
 $u = -\frac{g}{B} \Rightarrow \begin{pmatrix} -1 \\ -1 \end{pmatrix} \in \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
 $\Rightarrow \frac{1}{2} \leq \frac{1}{2} \checkmark$
 $\Rightarrow u^* = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$

Problem: $B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \Delta = \frac{1}{2}$
primal feasibility $u \in \Delta \Rightarrow u \in \mathbb{R}^2$
stationary $\nabla_u L = 0 \Rightarrow u^T B u = g$

Weak duality complementary $\lambda(u+1)^T B u = 0$
 $L(u, \lambda) = g^T u + \frac{1}{2} u^T B u + \lambda(u+1)^T B u$
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 $\Rightarrow \frac{1}{2} \leq \frac{1}{2} \checkmark$
 $\Rightarrow u^* = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$

Problem 4: $B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \Delta = \frac{1}{2}$
primal feasibility $u \in \Delta \Rightarrow u \in \mathbb{R}^2$
stationary $\nabla_u L = 0 \Rightarrow u^T B u = g$
Weak duality complementary $\lambda(u+1)^T B u = 0$
 $L(u, \lambda) = g^T u + \frac{1}{2} u^T B u + \lambda(u+1)^T B u$
Consider $\lambda = 0$
 $u = -\frac{g}{B} \Rightarrow \begin{pmatrix} -1 \\ -1 \end{pmatrix} \in \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
 $\Rightarrow \frac{1}{2} \leq \frac{1}{2} \checkmark$
 $\Rightarrow u^* = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$

Problem 5: $B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \Delta = \frac{1}{2}$
primal feasibility $u \in \Delta \Rightarrow u \in \mathbb{R}^2$
stationary $\nabla_u L = 0 \Rightarrow u^T B u = g$

Weak duality complementary $\lambda(u+1)^T B u = 0$
 $L(u, \lambda) = g^T u + \frac{1}{2} u^T B u + \lambda(u+1)^T B u$
Consider $\lambda = 0$
 $u = -\frac{g}{B} \Rightarrow \begin{pmatrix} -1 \\ -1 \end{pmatrix} \in \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
 $\Rightarrow \frac{1}{2} \leq \frac{1}{2} \checkmark$
 $\Rightarrow u^* = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$

Problem 6: $B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \Delta = 1$
primal feasibility $u \in \Delta \Rightarrow u \in \mathbb{R}^2$
stationary $\nabla_u L = 0 \Rightarrow u^T B u = g$
Weak duality complementary $\lambda(u+1)^T B u = 0$
 $L(u, \lambda) = g^T u + \frac{1}{2} u^T B u + \lambda(u+1)^T B u$
Consider $\lambda = 0$
 $u = -\frac{g}{B} \Rightarrow \begin{pmatrix} -1 \\ -1 \end{pmatrix} \in \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
 $\Rightarrow \frac{1}{2} \leq \frac{1}{2} \checkmark$
 $\Rightarrow u^* = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$

Problem 7: $B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \Delta = 1$
primal feasibility $u \in \Delta \Rightarrow u \in \mathbb{R}^2$
stationary $\nabla_u L = 0 \Rightarrow u^T B u = g$
Weak duality complementary $\lambda(u+1)^T B u = 0$
 $L(u, \lambda) = g^T u + \frac{1}{2} u^T B u + \lambda(u+1)^T B u$
Consider $\lambda = 0$
 $u = -\frac{g}{B} \Rightarrow \begin{pmatrix} -1 \\ -1 \end{pmatrix} \in \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
 $\Rightarrow \frac{1}{2} \leq \frac{1}{2} \checkmark$
 $\Rightarrow u^* = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$

Problem 8: $B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \Delta = 1$
primal feasibility $u \in \Delta \Rightarrow u \in \mathbb{R}^2$
stationary $\nabla_u L = 0 \Rightarrow u^T B u = g$
Weak duality complementary $\lambda(u+1)^T B u = 0$
 $L(u, \lambda) = g^T u + \frac{1}{2} u^T B u + \lambda(u+1)^T B u$
Consider $\lambda = 0$
 $u = -\frac{g}{B} \Rightarrow \begin{pmatrix} -1 \\ -1 \end{pmatrix} \in \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
 $\Rightarrow \frac{1}{2} \leq \frac{1}{2} \checkmark$
 $\Rightarrow u^* = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$

Problem 9: $B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \Delta = 1$
primal feasibility $u \in \Delta \Rightarrow u \in \mathbb{R}^2$
stationary $\nabla_u L = 0 \Rightarrow u^T B u = g$
Weak duality complementary $\lambda(u+1)^T B u = 0$
 $L(u, \lambda) = g^T u + \frac{1}{2} u^T B u + \lambda(u+1)^T B u$
Consider $\lambda = 0$
 $u = -\frac{g}{B} \Rightarrow \begin{pmatrix} -1 \\ -1 \end{pmatrix} \in \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
 $\Rightarrow \frac{1}{2} \leq \frac{1}{2} \checkmark$
 $\Rightarrow u^* = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$

Problem 10: $B = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, g = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \Delta = 1$
primal feasibility $u \in \Delta \Rightarrow u \in \mathbb{R}^2$
stationary $\nabla_u L = 0 \Rightarrow u^T B u = g$
Weak duality complementary $\lambda(u+1)^T B u = 0$
 $L(u, \lambda) = g^T u + \frac{1}{2} u^T B u + \lambda(u+1)^T B u$
Consider $\lambda = 0$
 $u = -\frac{g}{B} \Rightarrow \begin{pmatrix} -1 \\ -1 \end{pmatrix} \in \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
 $\Rightarrow \frac{1}{2} \leq \frac{1}{2} \checkmark$
 $\Rightarrow u^* = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$